Bihar Engineering University, Patna

B.Tech 4th Semester Examination, 2024

Course: B.Tech
Code: 103404

Subject: Signals and Systems

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven questions only):- **[2 x 7 = 14]**

(a) If X(f) is the Fourier transform of x(t), then what is the Fourier transform of x(2t)?

(i) $X(f/2)$ (iii) 2X(2f)
(ii) $0.5X(f/2)$ (iv) X(f)

(b) Which of the following statements is true?

(i) Discrete-time signals are always periodic (iii) Discrete-time signals can be periodic or aperiodic
(ii) Continuous-time signals cannot be periodic (iv) Continuous-time signals are always aperiodic

(c) Find the fundamental period of $x(t) = \sin(5\pi t) + \cos(7\pi t)$.

(i) 0.2 sec (iii) 0.8 sec
(ii) 0.4 sec (iv) 1.0 sec

(d) If a system's impulse response $h(t) = \delta(t) - \delta(t-1)$, the system is:

(i) Causal (iii) Non-causal
(ii) Anti-causal (iv) Time-invariant

(e) Which system property ensures that the output depends only on past and present inputs?

(i) Linearity (iii) Stability
(ii) Causality (iv) Time-invariance

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(f) Which of the following is an example of a deterministic signal?

(i) White noise (iii) Random signal
(ii) Sinusoidal signal (iv) Gaussian noise

(g) The impulse response of an LTI system is given by $h(t) = e^{-2t}u(t)$. Find the system response to an input $x(t) = e^{-t}u(t)$.

(i) $\frac{e^{-t} - e^{-2t}}{3}$ (iii) $\frac{e^{-t} - e^{-2t}}{2}$
(ii) $\frac{e^{-t} - e^{-2t}}{2}$ (iv) e^{-3t}

(h) The step response of an LTI system is obtained by:

(i) Taking the derivative of the impulse response (iii) Taking the Fourier transform of the impulse response
(ii) Integrating the impulse response (iv) Adding two impulse responses

(i) The Laplace transform of $e^{-at}u(t)$ is:

(i) $\frac{1}{s+a}$ (iii) $\frac{s}{s+a}$
(ii) $\frac{1}{s-a}$ (iv) $\frac{s}{s-a}$

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(j) If a signal contains frequency components up to 5 kHz, what should be the minimum sampling rate?

(i) 2.5 kHz
(ii) 5 kHz
(iii) 10 kHz
(iv) 20 kHz

Q.2 (a) Find the inverse Laplace transform of: $F(s) = \frac{s(s+2)}{(s+1)(s+3)(s^2+5)}$. [7]

(b) Given a system with a difference equation $y[n]-0.5y[n-1] = x[n]$. determine its stability. [7]

Q.3 (a) Find the state space model of the system represented by the differential equation: $\frac{d^3y}{dt^3} + 8\frac{d^2y}{dt^2} + 9\frac{dy}{dt} = 3x(t)$. [7]

(b) A system is described by the equation: $y(t) = 3x(t) + 2$. Check whether the system is linear or not. [7]

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Q.4 (a) Compute the poles and zeros of $F(s) = \frac{s+1}{s^3+7s^2+10s+18}$. [7]

(b) Consider a system defined by $y(t) = tx(t)$. Determine whether it is linear and shift-invariant. [7]

Q.5 (a) Compute the output y[n] of an LTI system with impulse response h[n]={1,2,1} and input x[n]={2,1,3}using convolution. [7]

(b) Compute the energy and power of the signal $x(t) = e^{-t}u(t)$. [7]

Q.6 (a) Find the Fourier transform of $x(t) = e^{-2|t|}$. [7]

(b) Find the Laplace Transform of $x(t) = e^{-t}u(t)$ and determine its region of convergence. https://www.akubihar.com [7]

Q.7 (a) Given x(t) = sin(10t) + cos(15t), determine whether it is periodic and find the fundamental period. [7]

(b) Compute the z-transform of $x[n] = -b^n u[n-1]$. [7]

Q.8 (a) Find the step response of a system with impulse response $h(t) = e^{-3t}u(t)$. [7]

(b) Determine the even and odd components of the signal $x(t) = t^2 + 3t + 2$. [7]

Q.9 Write short notes on ***any two*** of the following:- [7x2=14]

(a) Discuss the properties of a system with respect to linearity, causality, and stability.

(b) Define Fourier series and explain its importance in signal representation.

(c) State and explain the Sampling Theorem and its significance.

(d) State and explain Parseval's Theorem with its application.

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